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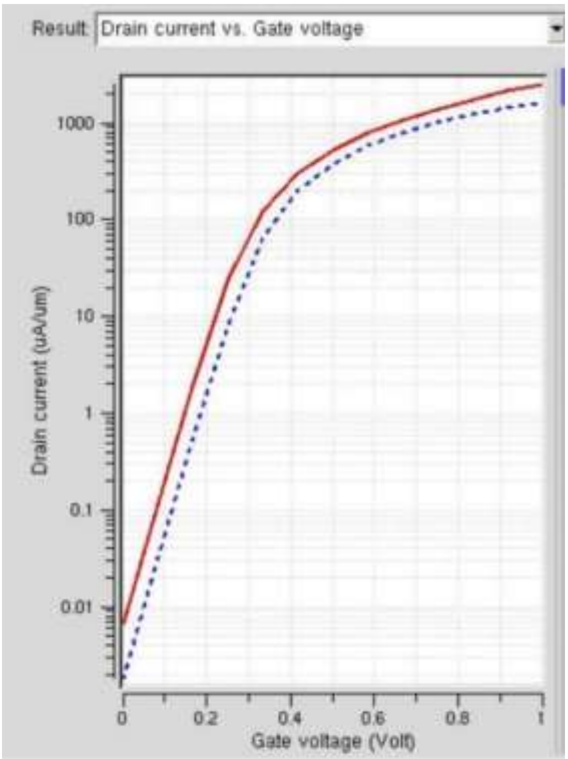
Nano electronics

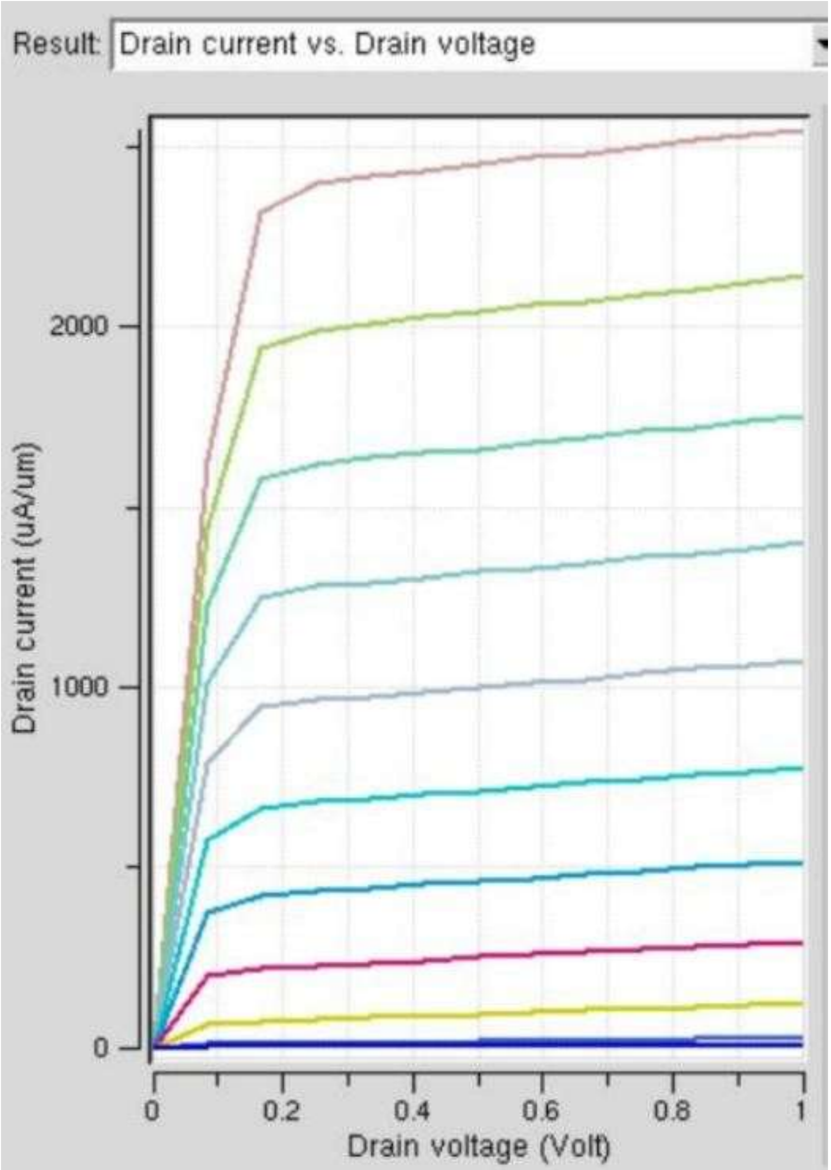
Exp6: MOSFET Scaling Analysis and Subthreshold Swing Calculation using

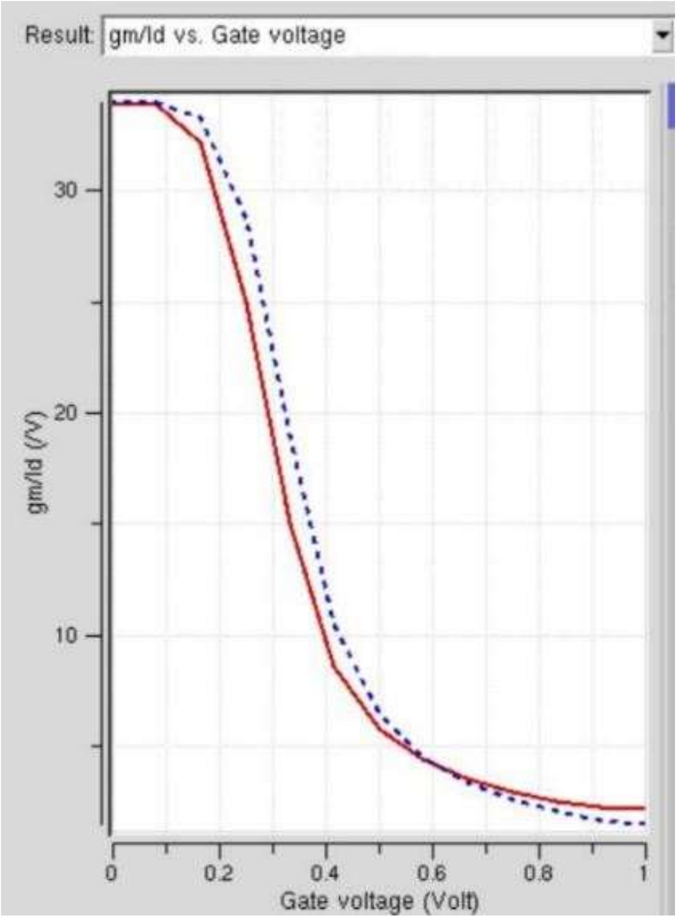
NanoHub

Output:



Case 2: Variation in Vd:





Outcome justification:
figure 1:

- At **low gate voltages**, the drain current is extremely small because the MOSFET is in the **OFF state**.
- Near the **threshold voltage (V_{TH})**, the drain current rises sharply, indicating channel formation.
- Beyond the threshold voltage, the MOSFET enters the **ON state**, where the drain current increases significantly.
- The **solid red curve** exhibits a slightly higher drain current than the **dashed blue curve**, indicating improved current drive and better switching performance.

Figure 2

At **small drain voltages**, the drain current increases almost linearly with drain voltage, indicating the **linear (ohmic) region**.

As the drain voltage increases further, the curves become nearly flat, indicating the **saturation region**.

Each curve corresponds to a different gate voltage.

Higher gate voltages produce higher drain currents because a stronger inversion channel is formed.

The slight increase in current after saturation is due to **channel length modulation**, which is commonly observed in practical MOSFETs.

The output characteristics verify that the MOSFET successfully transitions from the linear region to the saturation region and demonstrate the influence of gate voltage on drain current, confirming proper device operation.

- Figure3: At **low gate voltages**, the **g_m/I_D** ratio is high, indicating excellent current efficiency in the **subthreshold region**.
- As the gate voltage increases, the ratio gradually decreases because the MOSFET moves into the **strong inversion region**.
- The solid and dashed curves follow similar trends, indicating consistent device behavior with slight differences due to scaling or device parameters.

The graph demonstrates the trade-off between **current efficiency** and **drive current**. High **g_m/I_D** values are desirable for low-power circuits, while lower values correspond to high-performance operation.